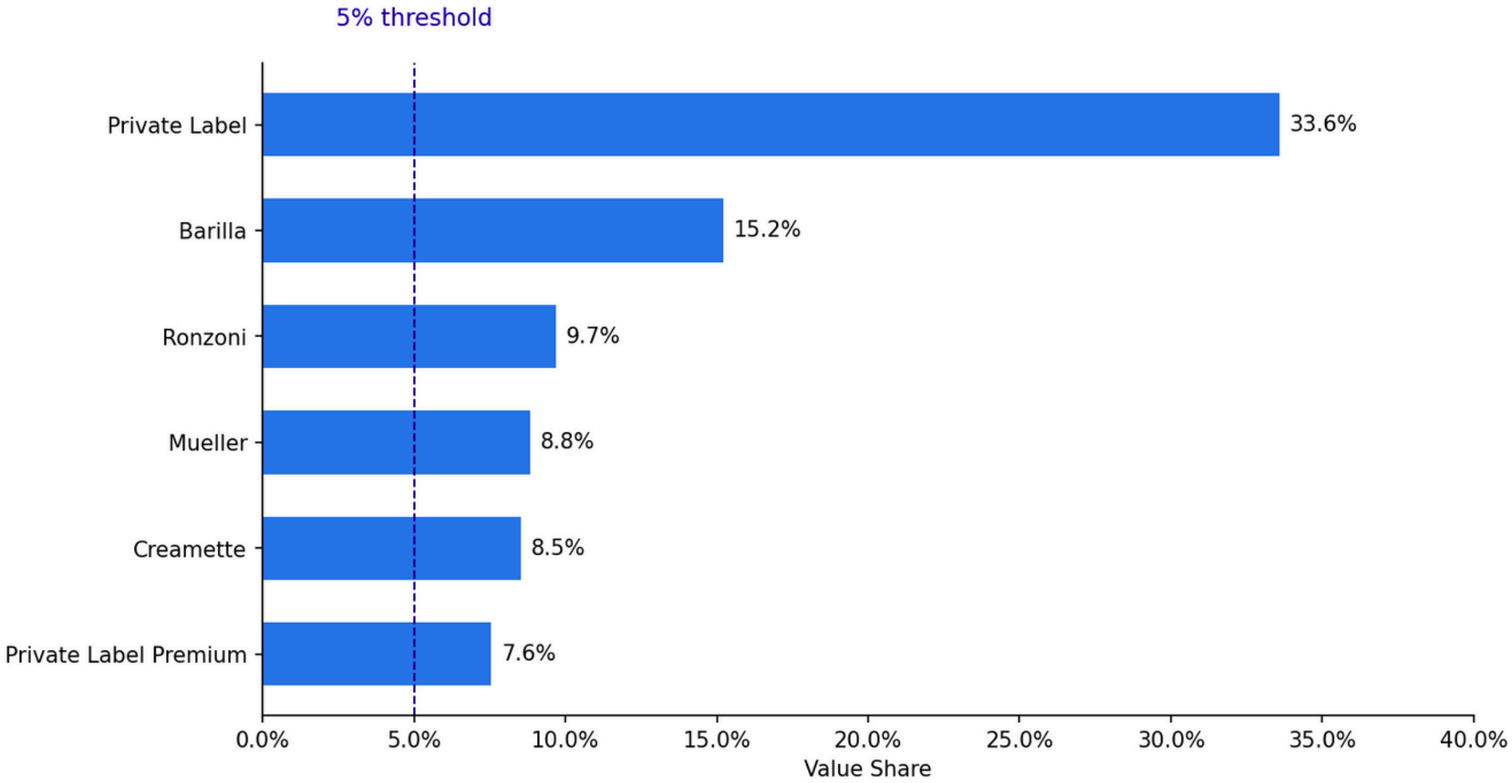
Dunnhumby Carboload PASTA Analysis

Pasta Category Brand Level Summary

	brand	Total_Revenue	Total_Profit	Share_of_Profit	Active_Buyers	Order_Frequency
0	Private Label	995288	995288	34.9%	65.0%	3.74
1	Barilla	433894	433894	15.2%	30.5%	2.57
2	Creamette	271595	271595	9.5%	21.1%	2.43
3	Mueller	270850	270850	9.5%	22.1%	2.23
4	Ronzoni	213727	213727	7.5%	16.7%	2.21
5	Private Label Premium	208820	208820	7.3%	18.6%	2.25
6	San Giorgio	124221	124221	4.4%	10.3%	1.99
7	Hodgson Mills	80522	80522	2.8%	4.6%	2.08
8	No Yolks	66296	66296	2.3%	6.8%	1.68
9	Private Label Value	41128	41128	1.4%	5.1%	1.49

The pasta dataset was cleaned and aligned by trimming incomplete weeks, removing unused fields, filling missing values, filtering out invalid sales, and restricting the data to pasta products only. The resulting dataset was then used for brand-level analysis.

Brand value Share (Pasta)



Brands with a market value share greater than 5% were selected as the competitive set. **Barilla** shows strong performance with 15% profit share—standing out as the top manufacturer brand against Private Label's broad dominance. As **the focal brand** in this analysis, Barilla's results provide the primary lens through which competitive dynamics in the pasta category are evaluated.

Focal Brand Customer Dashboard

Barilla Brand Dynamics (Q1–Q8)

		Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Total customers		125508	125508	125508	125508	125508	125508	125508	125508
Sales Metrics									
Total revenue		42392.56	55272.11	54697.38	51998.82	55317.36	59965.2	61039.52	53210.89
Total items		36426	53507	51428	48906	47282	56621	54191	45359
Total SKUs (cross-buying)		33	33	33	37	34	34	34	34
Buying metrics									
Total buyers		20533	28391	28103	25344	26698	29143	29732	25179
Total purchase occasions		25920	36752	36996	32868	35094	38182	39001	32379
Buyer penetrations (b)		0.16	0.23	0.22	0.20	0.21	0.23	0.24	0.20
Buyer purchase frequency (w)		1.26	1.29	1.32	1.3	1.31	1.31	1.31	1.29
Customer base health									
Purchase av. revenue ($)		1.64	1.5	1.48	1.58	1.58	1.57	1.57	1.64
Buyer av. revenue ($)		2.06	1.95	1.95	2.05	2.07	2.06	2.05	2.11
Q-Q repeat buyers	N/A		31%	28%	26%	28%	29%	29%	25%
New customers	N/A		21983	20032	18034	19589	21268	21366	17673

Focal Brand Customer Dashboard

Summary of Barilla Brand Dynamics (Q1–Q8)

The dataset covers 125,508 unique households who purchased pasta **at least once across the eight quarters**, forming the denominator for penetration and frequency metrics.

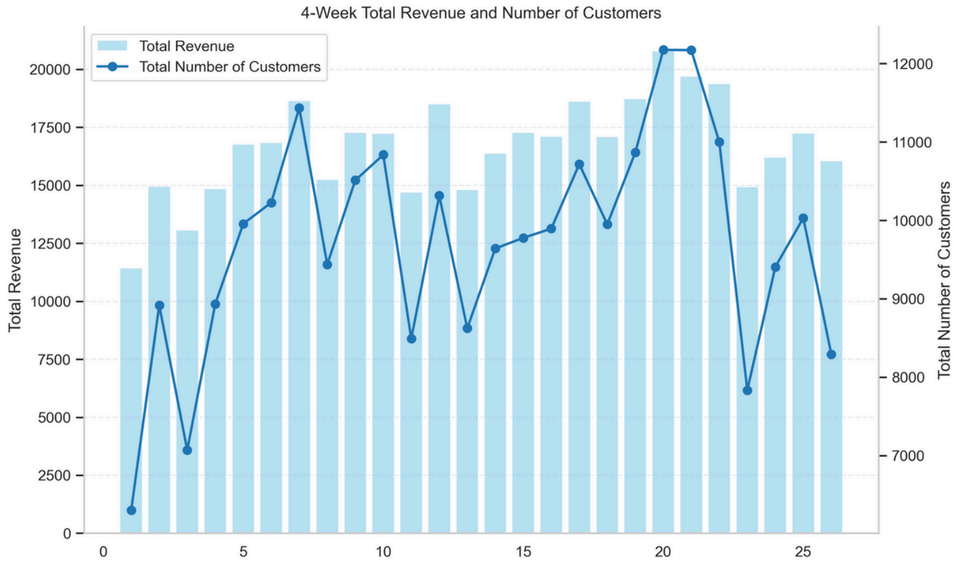
Category revenue fluctuates between $42.4K (Q1) and $61.0K (Q7), reflecting seasonal variations in demand. The number of total buyers ranges from ~20.5K in Q1 to ~29.7K in Q7, with corresponding buyer **penetration (b) between 16% and 24%.** Buyer **purchase frequency (w) remains stable** at around 1.3 purchases per buyer per quarter, indicating limited variation in purchase intensity.

Customer base renewal is evident: each quarter, 18K–22K new buyers enter the category, while quarter-to-quarter repeat buyers account for 25–31% of households. This indicates **substantial levels of new buyer acquisition alongside moderate levels of buyer retention.** Overall, the pasta category achieves broad household reach, with purchasing that is steady in occurrence but not characterized by high frequency.

Focal Brand Customer Dashboard

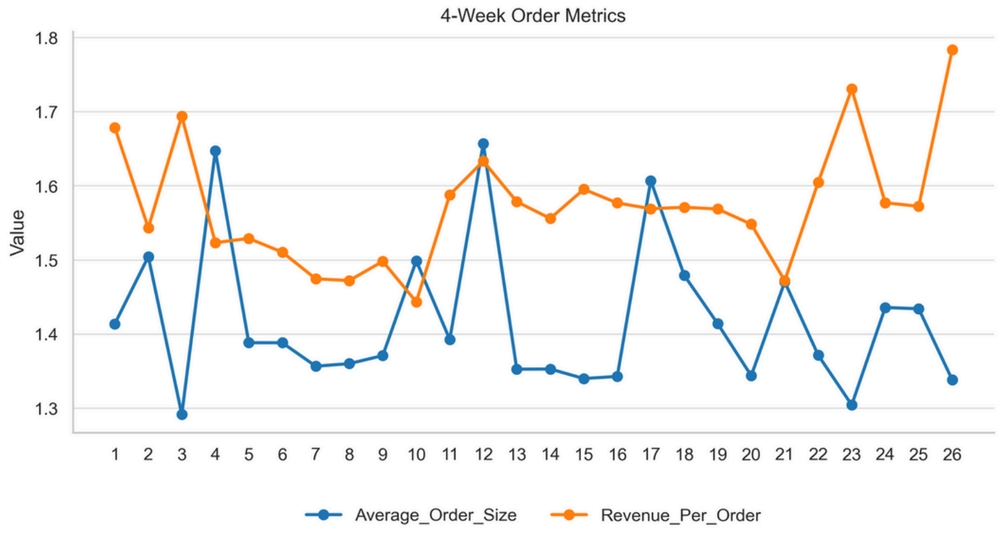
Barilla: 4-Week Total Revenue & Customers

- Revenue and customer counts move largely in tandem, with peaks in periods ~20–21 and troughs around ~11 and ~23.
- Given stable order sizes, revenue peaks coincide with higher customer counts, highlighting buyer recruitment and reach as the key growth driver.
- Seasonal or promotional cycles appear evident, suggesting different levers: scale reach during high-traffic windows and optimize basket value in lower-traffic periods.



Barilla: 4-Week Order Metrics

- Average order size remains stable at ~1.3–1.6 units, with only a few spikes (e.g., periods 4, 12, 17, 24–26).
- Revenue per order shows a mild upward trend toward the end, diverging from average order size.
- The weak correlation suggests that higher order revenue is driven more by price/mix effects (premium SKUs, promotions, pricing) than by customers buying more items per trip.



Implication: Barilla's revenue growth is driven mainly by customer acquisition and pricing/mix effects, while both purchase frequency per buyer and units per order remain relatively stable.

Focal Brand Customer Dashboard

Analysis of Buyers by buyer class for Quarters 1 - 8

r	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
0	104975	97117	97405	100164	98810	96365	95776	100329
1	16956	22920	22388	20449	21273	23328	23765	20479
2	2490	3777	3921	3356	3694	3988	4093	3272
3	666	1043	1063	928	1058	1078	1120	862
4	259	390	398	369	366	414	415	322
5	92	125	177	127	173	201	175	119
6	36	74	72	59	68	61	84	66
7	20	31	45	32	33	29	43	24
8	6	12	18	11	21	15	8	16
9	3	10	10	3	6	14	15	8
10+	5	9	11	10	6	15	14	11
Total buyers	20533	28391	28103	25344	26698	29143	29732	25179
b	0.16	0.23	0.22	0.2	0.21	0.23	0.24	0.2
w	1.26	1.29	1.32	1.3	1.31	1.31	1.31	1.29

Buyer distribution follows the **NBD** pattern, with many light buyers and few heavy buyers. .
Across quarters, the average purchase frequencies remain relatively stable around 1.3. This stability illustrates **regression to the mean**: extreme buyer classes (very heavy or very light buyers) tend to moderate toward the long-run category average over time.
Penetration (b = 0.16–0.24) varies more than frequency (w ≈ 1.3), indicating growth is mainly driven **by attracting new buyers rather than deepening loyalty**.

Focal Brand Customer Dashboard

Buyer behavior for Barilla in the Pasta category

Number of purchases	Category buyers %	Sales volume %	Share of requirement %
0	69.5		
1	18.3	26	42
2	5.5	16	38
3	2.5	11	39
4	1.4	8	41
5	0.8	6	43
6	0.5	5	45
7	0.4	4	47
8	0.3	3	48
9	0.2	3	50
10+	0.7	19	57

Light buyers, in aggregate, **drive most volume**
Households buying once contribute 26% of Barilla volume; 1–2 times = 42%; **1–3 times = 53%.**
▸Growth cannot rely on a small set of heavy users; getting more households to buy just one more time delivers bigger gains.

Heavy buyers are few but important — and **not "loyal only"**.
10+ times buyers are only 0.7% of households yet deliver 19% of volume; however their SoR is just 57% — 43% of their category purchases still go to other brands.
▸Loyalty rises with buying frequency (SoR climbs from 42% at one purchase to 57% at 10+), but it plateaus, so growth shouldn't be pinned on "squeezing a bit more" from heavies.

Focal Brand Customer Dashboard

Barilla Loyalty Metrics

Brand	Barilla
Brand Share	13.7%
Penetration	30.5%
% Buying Once	60%
% Buying 5+	10%
Repeat Buying	40%
Purchases per buyer (Brand)	2.2
Purchases per buyer (Category)	8
Share of Category Requirements	41%
100% Loyal - %	16%
100% Loyal – Rate	1.4

Full period: 104 weeks / 8 quarters
Barilla captures 13.7% category share with 30.5% penetration, and ***most buyers are light or occasional*** *(60% buy once, only 10% buy 5+).*
Loyalty is bounded*: repeat buying is 40% and SCR is 41%, while just 16% of households are 100% loyal. This pattern is consistent with a penetration-led brand, where growth depends on attracting many occasional buyers rather than cultivating heavy loyals.*

Competitive Performance

Pasta Brand Metrics – 8 Quarters

Pasta 8 Quarters	Market Share	Penetration	Purchase Frequency	Percent Buying Once	100% LOYAL		P. freq of the category
					pen.	p. freq.	
ALL	100	100	4.8	35	100	4.8	4.8
Private Label	44	65	3.4	44	22	1.9	6.0
Barilla	14	31	2.2	60	5	1.4	8.0
Creamette	9	21	2.3	57	4	1.5	7.8
Mueller	9	22	2.1	63	6	1.6	6.9
Private Label Premium	7	19	1.9	66	3	1.3	8.3
Ronzoni	7	17	2.0	63	2	1.3	7.9
Others	11	27	2.2	59	4	1.4	8.3
Average	14	29	2.3	59	7	1.5	7.6

Competitive Performance

Summary of Pasta Brand Metrics – 8 Quarters

Share tracks penetration. Brands with more buyers have more share; ***repeat rates sit in a narrow band****.*

Private Label leads on both share and reach; advantage is ***scale of buyers****, not unusually high frequency.*

Branded followers (Barilla, Creamette, Mueller, etc.) cluster with similar purchase frequency; ***differences are mainly penetration-driven****.*

Repertoire behaviour*: high % buying once and low 100% loyal % show shoppers rotate across brands.*

Category frequency among a brand's buyers is much higher than the brand's own frequency → these buyers are heavy category users, ***not single-brand loyalists****.*

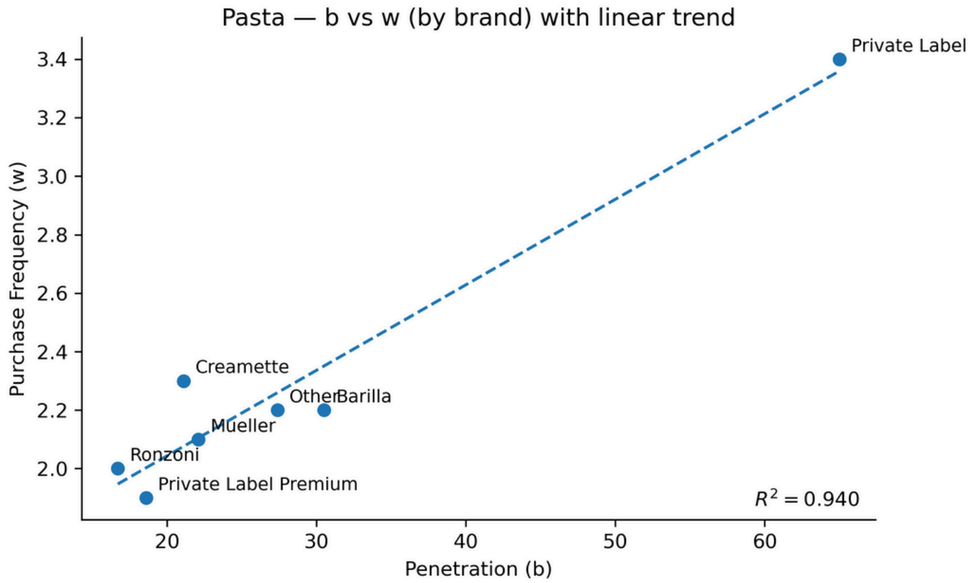
Implication: to grow share, ***recruit buyers*** *(distribution, availability, salience) first; then nudge repeat (activation, pack/price, promotions).*

Tail ("Others") still sizable → opportunity via assortment, value pack, or targeted activation.

Competitive Performance

Comparative Performance Metrics of Pasta Brands

		Share (%)	Penetration (%)	Purchase Frequency
Pasta				
	Private Label	44.1	65.0	3.4
	Barilla	13.7	30.5	2.2
	Creamette	8.8	21.1	2.3
	Mueller	8.7	22.1	2.1
	Private Label Pı	7.2	18.6	1.9
	Ronzoni	6.5	16.7	2.0
	Other	10.9	27.4	2.2



*Higher Market Share typically comes from **higher Penetration (b)***
*Smaller brands usually show lower b and w and a higher Percent Buying Once, consistent with the **Double Jeopardy** pattern.*

Competitive Performance

Data Coverage:	83%							
							Period 1	Period 2
	Private Label	Barilla	Creamette	Mueller	Private Label Premium	Ronzoni		
Private Label	55645	5581	5602	3295	2784	2812	75719	80294
Barilla	6541	7642	1333	1255	1441	1720	19932	19397
Creamette	5581	1332	6014	378	512	80	13897	13741
Mueller	6444	1877	198	7308	617	1758	18202	13585
Private Label Premium	3542	1536	583	465	2719	564	9409	8551
Ronzoni	2541	1429	11	884	478	2453	7796	9387
Total	80294	19397	13741	13585	8551	9387	144955	
							144955	

Split the most recent periods into two halves: ***Period 1 (P1) = earlier 13; Period 2 (P2) = later 13.***
This view lets us see retention (stayers), defection, and brand acquisitions between the two periods for the main brands

Recalculated totals for brands present representing 144955/174055 = ***83% of market****.*

Row totals *represent* ***Period 1*** *pasta brand buyers, with each brand's share calculated as the number of brand buyers divided by the total number of pasta category buyers.* ***Diagonal cells*** *indicate brand* ***stayers*** *across periods, while* ***defectors*** *are obtained by subtracting stayers from the Period 1 brand buyers.*

Observed vs Expected Brand-Switching under Duplication of Purchase Law

Brand	Period 1 Share	Period 2 Share	Stayers	Defectors	Repeat Rate
Private Label	52%	55%	55645	20074	73%
Barilla	14%	13%	7642	12290	38%
Creamette	10%	9%	6014	7883	43%
Mueller	13%	9%	7308	10894	40%
Private Label Premium	6%	6%	2719	6690	29%
Ronzoni	5%	6%	2453	5343	31%
SUM			81781	63174	
SWITCHING RATE				**44%**	
REPEAT RATE				**56%**	

Brand Defection

	Obs.	Th.	Dev.
	27%	31%	-5%
	62%	56%	**6%**
	57%	59%	-2%
	60%	57%	3%
	71%	61%	10%
	69%	62%	7%
Correlatio		**0.97**	

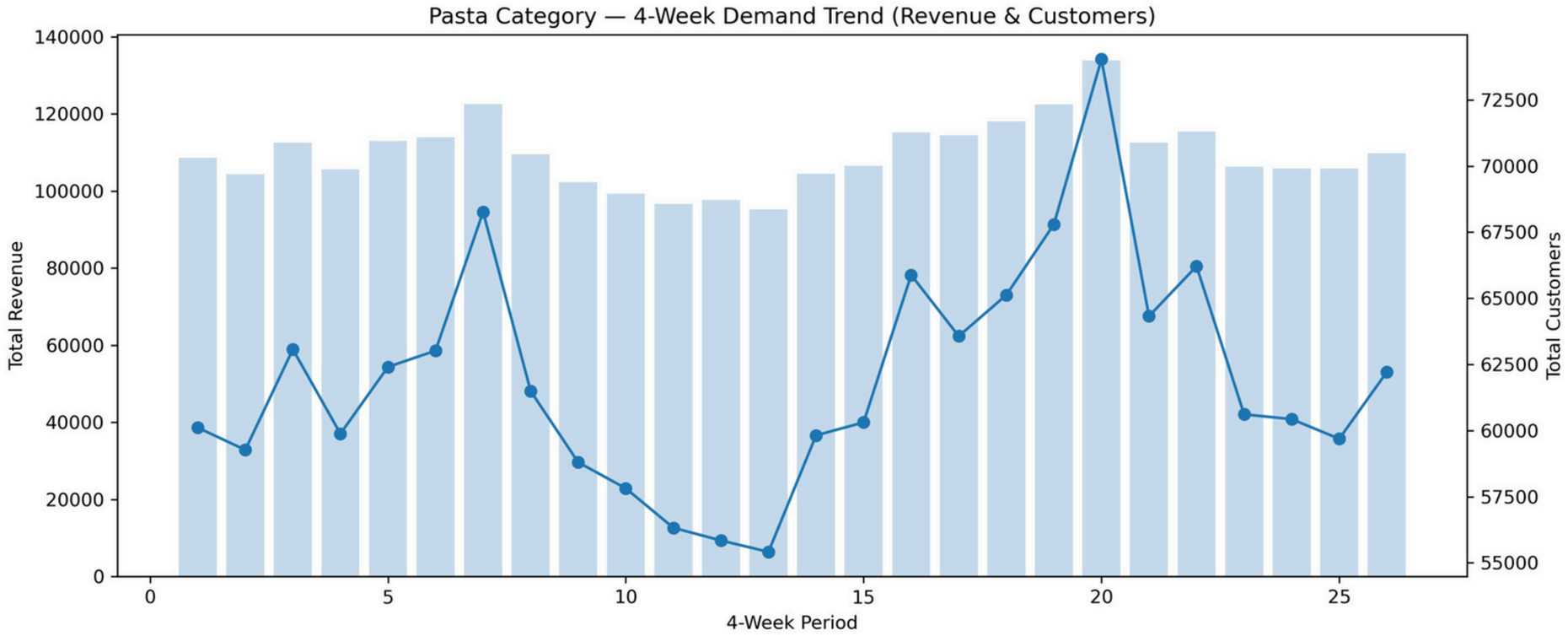
2nd Purchase

1st	Private Label	Barilla	Creamette	Mueller	Private Label Premium	Ronzoni
Private Label		-1194	878	-2892	-414	162
Barilla	-234		90	-374	599	1022
Creamette	857	89		-758	-75	-406
Mueller	257	248	-938		-152	1121
Private Label Premium	344	694	-4	-304		235
Ronzoni	-109	731	-475	247	149	

***Deviations** from theoretical expectations are **generally small**, indicating a good model fit (**correlation = 0.97**).*
***Barilla** shows a higher defection rate (+6%), which indicates **a relative weakness in retaining buyers** compared to expectations."*

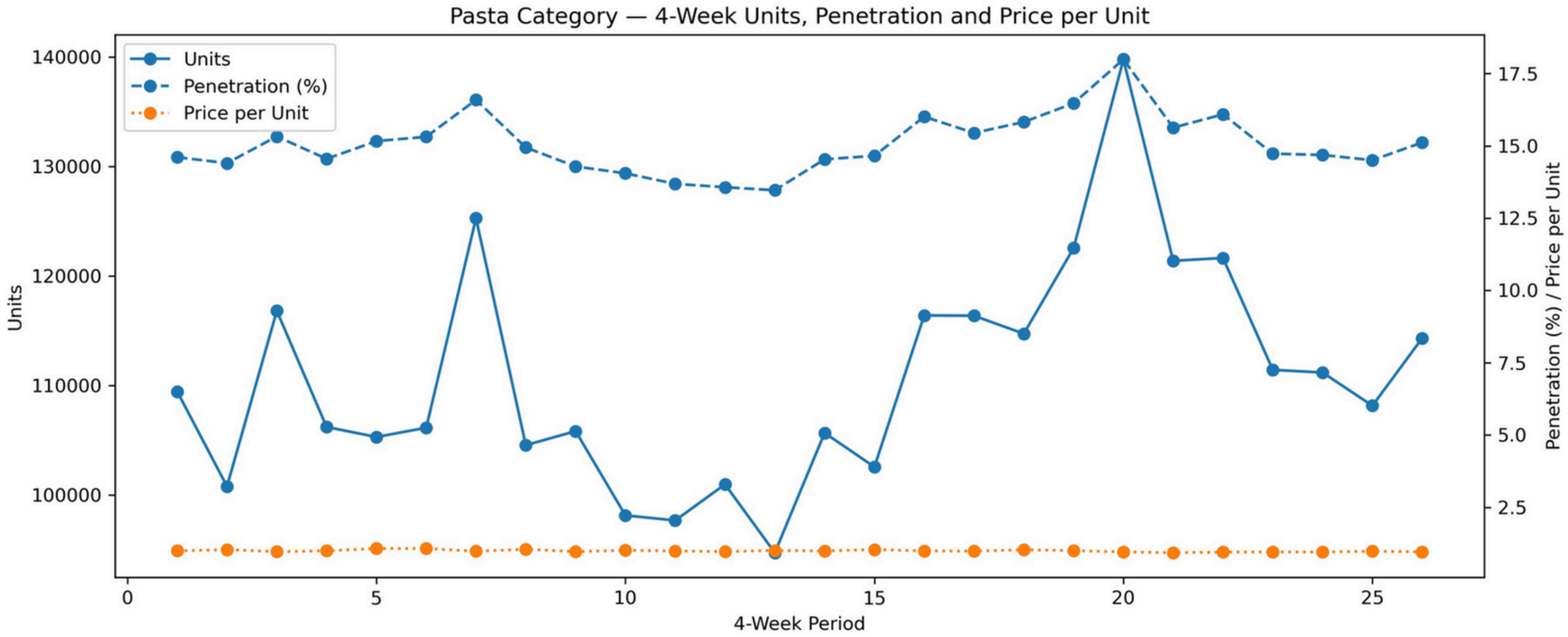
*This 2nd-purchase deviation table shows observed minus expected flows. For Barilla, inflows from **Private Label** are **below expectation (–1,194),** while inflows from other sources are above expectation. **The main shortfall into Barilla** therefore comes from Private Label buyers in period 2.*

Trends in Category Demand



*Category demand for pasta shows noticeable **seasonality**. Total revenue remains relatively stable overall but peaks sharply around periods 7 and 20, which coincide with higher customer counts. This indicates that **demand fluctuations** are driven primarily by changes in **the number of buyers** rather than shifts in purchase intensity.*

Trends in Category Demand



*The **penetration** curve co-moves with sales, confirming that these peaks are driven by fluctuations in the number of buyers. In contrast, the **unit price** remains **flat** throughout, suggesting price stability and **ruling out price effects** as a driver of demand variation.*

Last Quarter Performance/Expected

Purchase Class Distribution for Barilla in Q8

				Average per buyer		% of Total	
	Number_of_Buyers	Total_Revenue	Total_Items	Revenue	Items	Revenue	Customers
Purchase_Class							
0	100329						
1	20479	32514	28132	1.59	1.37	61.11%	81.33%
2	3272	11098	9265	3.39	2.83	20.86%	12.99%
3	862	4589	3760	5.32	4.36	8.62%	3.42%
4	322	2338	1920	7.26	5.96	4.39%	1.28%
5	119	1106	903	9.29	7.59	2.08%	0.47%
6	66	696	601	10.55	9.11	1.31%	0.26%
7	24	272	244	11.33	10.17	0.51%	0.10%
8	16	224	197	14.00	12.31	0.42%	0.06%
9	8	121	116	15.13	14.50	0.23%	0.03%
10+	11	252	221	22.91	20.09	0.47%	0.04%
Totals	125508	53210	45359	100.77	88.29	100.00%	100.00%
b	0.20						
w	1.29						

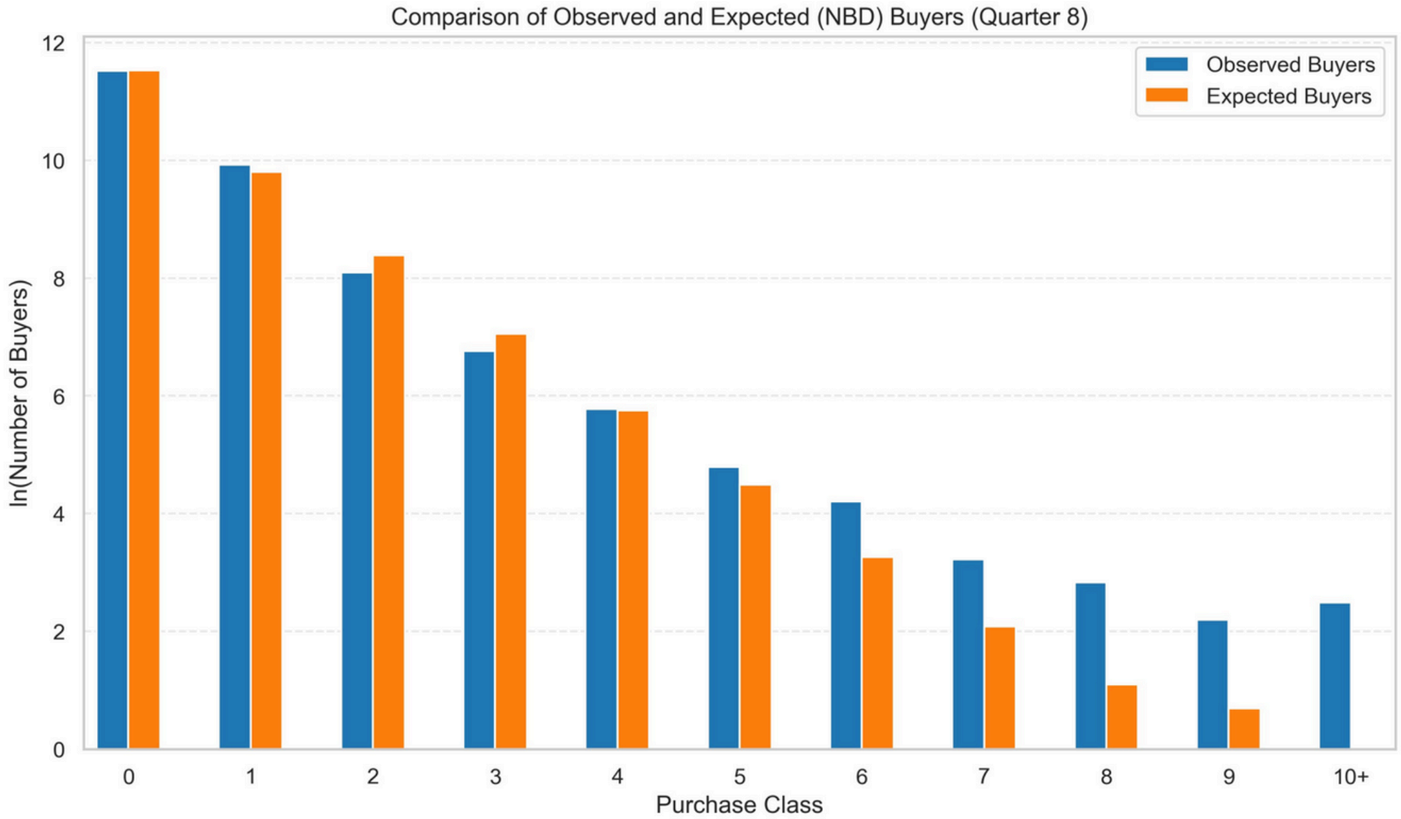
	Purchase_Class	Avg_Revenue	Avg_Items
Purchase_Class	1.000000	0.966412	0.975665
Avg_Revenue	0.966412	1.000000	0.996842
Avg_Items	0.975665	0.996842	1.000000

19% of customers contribute **~40% of the revenue.**
Purchase class is strongly correlated with average per buyer revenue and items, with **correlation** coefficients of approximately **0.97**

Last Quarter Performance/Expected

Barilla: Comparison of Observed and Expected Buyer Distributions (Q8, NBD Model)

Purchase Class	Observed Buyers	Expected Buyers
0	100329	101414
1	20479	18126
2	3272	4380
3	862	1151
4	322	314
5	119	88
6	66	25
7	24	7
8	16	2
9	8	1
10+	11	0
Total	125508	125508



The comparison between observed and expected buyer distributions **in the last quarter** shows a close alignment with the **Negative Binomial Distribution**. To improve visualization and make the scale more comparable across purchase classes, buyer counts were transformed using the **natural logarithm**. Most purchase classes, particularly one-time and two-time buyers, **follow theoretical expectations**. Observed buyers slightly exceed expectations in higher purchase classes (6+), but the overall fit remains strong with only minor tail deviations.

Last Quarter Performance/Expected

Last Q8

Brand	Market share %	Penetration b	Frequency w
Private Label	47	0.384	1.5
Barilla	13.5	0.129	1.29
Creamette	8.9	0.085	1.28
Mueller	8.8	0.085	1.26
Private Label Premiur	7.8	0.076	1.25
Ronzoni	8.8	0.087	1.25
Other	11.9	0.114	1.28
Average	15.2	0.137	1.3
Category Pasta	47	0.712	1.72

Q8 results illustrate the Double Jeopardy pattern: larger brands win mainly through higher penetration, while purchase frequency remains similar across brands. Category growth is driven by recruitment rather than repeat.

Last Quarter Performance/Expected

Brand	Market Share	Penetration (b)		Buying Once		Buying Five+		Frequency (Brand)		Frequency (Category)	
		O	T	O	T	O	T	O	T	O	T
Private Label	47	38.4	42.01	44.74	68.1	9.87	1.33	1.5	1.49	3.08	3.52
Barilla	13.5	12.9	14.26	60.37	81.52	5.15	0.41	1.29	1.25	3.55	3.82
Creamette	8.9	8.5	9.57	65.69	83.36	3.39	0.28	1.28	1.22	3.09	3.86
Mueller	8.8	8.5	9.59	68.09	83.38	2.48	0.25	1.26	1.22	2.81	3.82
Private Label Premium	7.8	7.6	8.52	61.1	83.82	4.57	0.21	1.25	1.21	3.71	3.85
Ronzoni	8.8	8.7	9.56	62.89	83.29	3.14	0.22	1.25	1.22	3.42	3.88

The comparison of observed versus theoretical values in the last quarter shows that brand performance **largely aligns with marketing law expectations.** Private Label leads with the widest penetration (38.4% observed vs. 42.0% expected) and above-average frequency, while Barilla ranks second (12.9% vs. 14.3%), maintaining moderate repeat levels. Mid-tier brands such as Creamette, Mueller, PL Premium, and Ronzoni all cluster around 8–9% share, with penetration and frequency close to theoretical predictions, confirming limited differentiation in repeat rates across brands. Differences in brand size are therefore primarily explained by penetration, consistent with the Double Jeopardy pattern.

Thank You